

The Pompeii Worm: Is Thermophily a Relic or Novelty?

Unlike modern seas, Archaean oceans were highly volatile, characterized by high hydrothermal activity, water temperatures, and concentrations of sulfur and other inorganic compounds [1]. Today, hydrothermal activity is less frequent and volatile, but like primitive oceans, undersea volcanic activity creates hypoxic, sulfuric, acidic, and hot conditions [1][5]. Despite this, organisms representing multiple phyla are highly adapted to these conditions, forming small, active communities of extremophiles [1][5]. Most notable are tube worms such as those in family Alvinellidae; five extant species are thermophilic [3], including *Alvinella pompejana*, the most thermotolerant metazoan on Earth [2][4][6].

A. pompejana is still a bit of a mystery; they build tubes directly upon active volcanic chimneys and are found at considerable depth [2], posing issues for studying *in situ* and *in vivo*. Examination of frozen tissue evidenced thermotolerant protein domains, adaptations for increased respiration and redox capacity, thermotolerant collagen, and an episymbiont of bacterial filaments that may also impart heat protection [2][3][6][7]. In addition to these adaptations, they persist temporally and spatially despite the volatile and ephemeral nature of vent activity [7]. Considering the resemblance of their ecosystem to primitive seas and the lineage of thermophily in prokaryotes [6], modern hydrothermal vent organisms raise the question of whether or not thermophily is the ancestral condition or an acquired trait – is that the case for *A. pompejana*?

As a trait, thermophily in prokaryotes has been studied at length and the fossil record shows evidence of microbial activity associated with hydrothermal vents [5]. Researchers found positive correlations between thermotolerance and certain hydrophobic amino acid residues, charged amino acid residues, and a high guanine-cytosine content in their structural RNAs [6]. A study in 2012 by Jollivet et al. [6] used this prokaryote criteria to compare the proteome and genome of *A. pompejana* to a close member of its family, *Paralvinella grasslei*, a cold-adapted tube worm (mesophile). Comparative analyses sought to quantify mutations associated with thermophily, and proteomes were compared to other lophotrochozoans; *A. pompejana* had the most charged aa residues and high hydrophobic aa residues [6]. Their phylogenetic reconstruction and positive selection modeling indicated that *P. grasslei* was likely under positive selection while *A. pompejana* was under negative selection, suggesting that high hydrophobic aa residues and charged residues were present in the common ancestor of the worms [6].

In a 2017 study, Fontanillas et al.[3] expanded upon the 2012 study and compared genomes and proteomes of 6 species of Alvinellid tube worms [3]. Two species of mesophiles and four species of thermophiles were sampled for genomic and proteomic sequencing. As in the 2012 study, researchers mapped proteomes, quantified and qualified aa composition, reconstructed an ancestral tree, and quantified and compared mutations in aa sequences. Using this data, they categorized worms based on thermophilic criteria for prokaryotes; cold-adapted worms and hot-adapted worms were clearly divided by temperature-associated criteria when plotted along three thermophily indices [3][4]. Aa residues were also different in “cold” worms versus “hot” worms, and aa residues in “hot” worms were associated with aa residues at ancestral nodes. Analysis of nonsynonymous mutation rates indicated that “cold” worms had a higher rate than “hot” worms; as in the 2012

study, researchers concluded thermophily was likely the ancestral state [3]. Also, if thermophily in *A. pompejana* was a “modern” development as a result of positive selective pressures, it would likely be associated with novel protein domains, but it is associated with the same or similar protein domains as thermophilic prokaryotes [3][4][7].

Ideal follow-up studies include additional genomic / proteomic sequencing among members of closely-related and distantly-related groups in order to resolve obscure or ambiguous phylogenies. Comparative analyses of genomes and proteomes using more resolved taxonomy and computer modeling may reveal a pattern of appearance or disappearance in phyla lineages, quantify and qualify selective pressures on thermophily, and establish timeframes of speciation events that may coincide with historical changes in atmospheric and geothermal conditions. Additionally, further exploration of deep-sea vent communities is vital to sampling more species and finding more *A. pompejana* colonies; bigger sample sizes and resolved phylogenetic relationships improve the validity and accuracy of theoretical models.

A currently-impossible-but-ideal study on thermophily is experimentation on live organisms. High pressure tanks of hot-adapted and cold-adapted Alvinellidae worms (and perhaps other vent annelids) are modified to experimental conditions of varying chemical and temperature gradients resembling all possible environmental conditions. Using varying conditions and applying slow and abrupt adjustments to conditions (then flash-freezing and examining worm tissues) may clarify theories regarding selection on thermal preference and how organisms from each category may respond to environmental change from their adapted condition.

Considering the difficulties in studying live organisms, current research is limited to molecular biology, evolutionary biochemistry, and genetics – all of which are providing invaluable comparative data, resolving many phylogenetic questions, and pointing toward thermophily as the ancestral condition. Technology for studying deep-sea organisms *in vivo* continues to improve, however, and eventually, experimentation on live worms may provide better insight into the question of ancestral vs derived thermophily in *A. pompejana*.

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